

Jeffrey Hoelzel Jr

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Work Experience

Software Engineer Intern | Altored Health | Flagstaff, AZ

Jan 2026 - Present

- Architected semantic retrieval workflows using pgvector and Meilisearch to match ambiguous patient queries with structured healthcare visit targets.
- Implemented confidence-gated LLM fallback logic with strict JSON outputs, increasing correct target selection from 67% to 98%.
- Dockerized backend services and search infrastructure to support local development, deployment testing, and scalable API orchestration.

Machine Learning Engineer | Pathogen and Microbiome Institute | Flagstaff, AZ

Aug 2025 - Present

- Developed PepSeqPred, a PyTorch-based machine learning (ML) pipeline to predict antibody epitope locations across diverse pathogens.
- Optimized ESM-2 embedding generation pipelines using batching and SLURM GPU parallelization, reducing high-performance compute (HPC) runtime by over 50%.
- Built logging and evaluation tools using PR AUC, recall, precision, F1, and confusion matrices to validate model performance under extreme class imbalance.

Software Engineer, Data Systems | NAU SICCS | Flagstaff, AZ

Jan 2025 - Present

- Engineered Python-based web scrapers with Selenium and BeautifulSoup to extract unstructured UltraSignUp trail race data for downstream statistical analysis.
- Improved scraping throughput by nearly 90% via concurrency and adaptive wait time strategies.
- Reduced OpenStreetMap API calls by 48.5% through dynamic GPX coordinate batching.

Information Technology Intern | Cavco Industries, Inc. | Phoenix, AZ

Jun 2025 - Aug 2025

- Built agentic LLM workflows to classify, route, and update ServiceDesk Plus ticket operations through automated tool calls.
- Designed a retrieval-augmented generation (RAG) pipeline, reducing response latency by 60% and consolidating 3+ API calls into 1 request.
- Developed an ML pipeline clustering 100,000+ tickets to automatically generate Confluence knowledge articles.

Software Engineer, Applied AI | CANIS Lab | Flagstaff, AZ

Jan 2025 - Jul 2025

- Built a campus health chatbot using React and Flask to support student health inquiries and facilitate research studies.
- Improved response relevance by 35% by refining RAG pipelines and contextual ranking.

Projects

ArtemiS3 | Capstone Project (USGS and NASA)

Svelte | FastAPI | PostgreSQL | Boto3 | Meilisearch

- Architected a Dockerized search platform enabling discovery across public NASA and USGS AWS S3 datasets.
- Built REST API endpoints with FastAPI and optimized queries using Meilisearch and Boto3.
- Led technical planning and stakeholder presentations, delivering architecture, feasibility, and requirements documentation.

Louie's Ratings | Software Engineering (CS386)

React | Flask | PostgreSQL

- Led a five-person team through sprint planning and delivery of a full-stack academic rating and grade insights platform.
- Developed secure authentication flows using React and Flask with bcrypt password hashing.
- Implemented unit and end-to-end acceptance testing with unittest and Selenium to ensure deployment stability.

Education

Northern Arizona University, Flagstaff, AZ

GPA: 3.85

B.S. in Software Engineering | Minor in Mathematics

Dean's List

Awards: Outstanding Senior Award | Keim Undergraduate Research Excellence Award, Honorable Mention | HURA Grant Recipient

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, C, Java, HTML, CSS

Libraries/Frameworks: React, Svelte, Node.js, FastAPI, PyTorch, Scikit-Learn, Pandas, NumPy

Databases/Search: PostgreSQL, pgvector, SQLite, Firebase Firestore, Meilisearch

Developer Tools: Docker, Docker Compose, Git, SLURM (Bash), AWS, Google Cloud, Azure, NGINX, HPC, CI/CD, Agile